Speed Testing

Prerequisites

1) The project is to be completed by **teams of two**.
2) You can use any programming language and platform you want. However, make sure your code is easy to read, with sufficient explanations and comments. Use a proper and consistent naming convention.
3) Do not simply copy/paste someone else's or AI code. You cannot simply use an existing library for the core functionality of speed testing.
4) You can discuss the project among yourselves or in the Google Classroom forum, but do not use each other's code.
5) You must enclose instructions (a readme file) on how to compile, build and/or run your code (along with any system requirements). Refrain from requiring too many things.
6) Start early and try out different ideas before the submission.

Submission and Demo

Create a single zip file that includes everything, name it lastnames_project1.zip and upload it to Google Classroom. We will have the demos during a designated class hour or office hours.

Problem to Solve

You will build a speed-testing tool on a website hosted in the cloud (AWS, Google, whatever your favorite is) that allows users to measure their upload and download speeds. Alternatively, if you have access to a Web server you know is not throttled, you can use it as well. Ideally, you will need objects of various sizes for proper testing.

- A user will visit your website (URL), and once the start button is clicked, the measurement will start. First, the ping time will be reported (the round-trip time between the user's browser and your server) in milliseconds. Then, the download speed will be measured and reported, and finally, the upload speed will be measured and reported. Do not forget the units.
- For an idea, you can visit https://www.speedtest.net/, https://fast.com/ or https://www.hiztesti.com.tr/. The course reading list also has some pointers.
- The UI does not have to be very fancy, but the test progress and results should be visible clearly.
- Your goal is to produce accurate results in the shortest time possible and with the least data transferred (for upload and download). In each test, you must report how much data was downloaded and uploaded by the user.
- To cross-check your results, compare them with those from other speed-testing tools on nearby servers. Your results may be a bit different, but they should not be too far from theirs.